# Organophosphate Poisoning

**Assess for Organophosphate Poisoning**

**Scene clues:**
- Presence of pesticide containers, sprayers, or farming equipment
- Chemical odor or unusual smell (often described as "garlic" or "petroleum")
- Multiple patients affected simultaneously
- Occurrence in agricultural, pest control, or industrial settings

**Patient history or witness report:**
- Recent use of insecticides, rodenticides, or nerve agents
- Intentional ingestion or exposure (e.g., suicide attempt)

**Signs of cholinergic toxicity:**
- **SLUDGE:** Salivation, Lacrimation, Urination, Defecation, Gastrointestinal distress: Nausea, vomiting, or abdominal cramping, Emesis.
- Muscle twitching, pinpoint pupils
- Respiratory symptoms such as coughing, wheezing, or secretions

**Scene Safety**
- Consider withdrawing from the scene and request Civil Defense for Decontamination
- Don Appropriate PPE

**Primary Survey**
Assess and Manage Airway, Breathing and Circulation as per Universal Care CPG

**Signs of Shock**

**Administer IV Fluid**
Adults: 250-500mL
Pediatrics: 10-20mL/kg

EMT

**Symptomatic Bradycardia,** or **Excessive bronchial Secretions** or **Respiratory Distress**

**Atropine IV**
**Adults: 2mg** initial dose.
Subsequent doses doubled every 3-5 minutes (4mg/6mg/8mg)

EMT-A

**Atropine IV**
**Pediatrics:**
**0.02-0.05mg/kg**
Repeat every 3-5 minutes

EMT-A

## Patient Care Goals

1. Rapid recognition of organophosphate toxicity and appropriate escalation
2. Protect EMS crew through use of PPE and safe scene practices
3. Prevent airway compromise and support oxygenation and ventilation
4. Administer atropine early in symptomatic patients (if indicated)
5. Prevent further toxin absorption by decontamination when safe
6. Ensure prompt ALS backup and transport
7. Provide accurate information to the receiving facility for targeted treatment

## Patient Inclusion & Exclusion Criteria

### Inclusion Criteria

- Suspected or confirmed exposure to organophosphates through:
  - Inhalation (spraying, fumigation, enclosed space)
  - Ingestion (intentional or accidental)
  - Dermal absorption (contact with liquid or powder)
  - Presence of cholinergic symptoms (e.g., salivation, vomiting, miosis, respiratory distress)
  - History of occupational or environmental exposure (e.g., agriculture, pest control, industrial chemicals)
  - Patient found in contaminated environment with suggestive signs (e.g., foam, bradycardia, secretions)
- **Signs, Symptoms and History consistent with Organophosphate Poisoning:**
  - Scene clues:
    - Presence of pesticide containers, sprayers, or farming equipment
    - Chemical odor or unusual smell (often described as “garlic” or “petroleum”)
    - Multiple patients affected simultaneously
    - Occurrence in agricultural, pest control, or industrial settings
  - Patient history or witness report:
    - Recent use of insecticides, rodenticides, or nerve agents
    - Intentional ingestion or exposure (e.g., suicide attempt)
  - Signs of cholinergic toxicity:
    - Excessive salivation, sweating, vomiting, diarrhea
    - Muscle twitching, pinpoint pupils
    - Respiratory symptoms such as coughing, wheezing, or secretions
- Any of these findings should trigger immediate PPE precautions, early ALS request, and preparation for airway compromise.

### Exclusion Criteria

- Exposure to a non-organophosphate toxin without signs of cholinergic toxicity
- Isolated skin or eye irritation without systemic symptoms
- Scene requiring HazMat-level decontamination or involving chemical warfare agents beyond EMS capabilities
- Cases where clear alternate diagnosis explains symptoms and no toxic exposure is suspected

## Key Patient Assessment Points

1. **Scene Assessment**
   - Is it an agricultural, industrial, or indoor spraying location?
   - Are there chemical containers, unusual smells, or multiple affected people?
   - Ensure crew PPE is worn before patient contact
   - General Observation
   - Excessive secretions: drooling, tearing, or vomiting
   - Restlessness, altered consciousness, seizures
   - Difficulty breathing or obvious respiratory fatigue
2. **Airway**
   - Listen for gurgling, crackles, or excessive secretions
   - Determine if patient is protecting their airway

- Suction may be needed immediately

3. **Breathing**
   - Respiratory rate: may be rapid early, then slow with fatigue
   - $SpO_2$: may drop due to secretions or bronchospasm
   - Auscultate for wheezing, wet lungs (bronchorrhea), silent chest
4. **Circulation**
   - Bradycardia is common (may be severe)
   - Hypotension may occur late or with co-ingestants
   - Capillary refill, pulse quality, skin perfusion
5. **Neurological Status**
   - GCS assessment: monitor for deterioration
   - Observe for fasciculations, tremors, seizures
   - Agitation may indicate hypoxia or early CNS toxicity
   - Pupils
     - Pinpoint (miosis) is a hallmark finding
     - Poor reactivity may accompany CNS depression
   - Skin
     - Sweating (diaphoresis) is usually profuse
     - Remove clothing and inspect for skin exposure or patches
   - Blood Glucose
     - Always check in altered mental status
     - Treat per DCAS Hypoglycemia or Hyperglycemia CPG if abnormal
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

## Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for:
     - Respiratory distress, bradycardia, or airway secretions are present
     - There is altered level of consciousness or seizures
     - Due to the large amount of Atropine required, consider requesting a 2nd ALS unit.
   - TeleEMS should be consulted if ALS unavailable.
1. **Scene Safety & Decontamination**
   - Wear full PPE: gloves, mask, eye protection; gown or coverall
   - Remove patient from source of exposure (e.g., out of sprayed field or enclosed space)
   - Remove contaminated clothing carefully and seal in a plastic bag
   - Irrigate exposed skin and eyes with clean water or saline if not yet decontaminated
   - Avoid contaminating ambulance or personnel
2. **Primary Survey**
   - Perform Primary Survey as per Universal Care CPG.
     - Suction airway to manage bronchorrhea and reduce aspiration risk
     - Administer high-flow oxygen via non-rebreather mask or BVM
     - If patient is not breathing adequately, begin assisted ventilation with BVM
     - Prepare for rapid respiratory failure; have airway equipment ready
     - Position patient in lateral recovery if unresponsive or vomiting
   - Monitor heart rate and rhythm closely for bradycardia or blocks, perform ECG and serial ECG if patient is compliant
3. **IV Access**
   - Establish IV access.
   - Consider IO access if IV access is not readily available.

4. **Fluid Administration**
   - Administer 0.9% Sodium Chloride (normal saline) if hypotension develops
     - Adults: 250-500mL
     - Pediatrics: 10–20 mL/kg IV
     - Reassess after each bolus for improvement in perfusion
   - Avoid fluid overload, especially if signs of pulmonary edema are present
5. **Atropine Administration**
   - Indication: Symptomatic bradycardia, bronchial secretions, or respiratory distress
   - Titrate to effect (chest clear of secretions), not heart rate; the primary goal is drying secretions and improving ventilation.
   - Do not delay atropine administration if the patient has respiratory symptoms
   - Dose:
     - Adult:
       - 1st Dose: 2mg IV/IO/IM
       - Subsequent Doses: 4mg/8mg/16mg IV/IO/IM
     - Pediatric:
       - 0.02 mg/kg IV/IO/IM
       - Minimum dose: 0.1 mg
       - Maximum single dose: 0.5 mg
   - Repeat doses every 5 minutes as needed until clinical improvement is observed
   - Atropine doses will likely exceed the amount of atropine available in the ambulance. Prioritize transport.
2. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Airway patency and presence of secretions
     - Heart rate and blood pressure
     - Pupil size and reactivity
     - Response to atropine; drying of secretions, improvement in breathing
     - Seizure activity, fasciculations, or muscle weakness

## Patient Safety Considerations

- Always wear gloves, mask, and eye protection; use gowns when available
- Do not delay atropine if symptoms are present; treat per clinical effect, not vitals alone
- Avoid direct skin contact with the patient's clothing or vomit
- Ensure suction and BVM are ready; anticipate airway deterioration
- Secure and label contaminated clothing or material for hospital handover
- Monitor for re-narcotization-like recurrence; symptoms may return if exposure persists
- Use Broselow tape for all pediatric dosing, including atropine

## Clinical Insights & Additional Guidance

- Respiratory failure is the primary cause of death in organophosphate poisoning; act early
- Bradycardia is common but not always the best indicator; focus on secretions and ventilation
- Atropine should be titrated until the lungs are dry, not based solely on HR
- Muscle weakness and paralysis (nicotinic effects) are not reversed by atropine
- Pediatric exposures can occur from crawling in treated areas or touching contaminated surfaces
- Never withhold transport in a symptomatic patient; even mild symptoms can worsen rapidly
- Consider mass exposure and notify authorities early if multiple cases arise

## Documentation

- Identification: Patient's name, age, sex, and weight (especially for pediatric dosing)
- Medical complaint: Suspected or confirmed organophosphate poisoning

- Information: Time and route of exposure, agent if known, amount, scene description
- Signs: GCS, HR, RR, BP, $SpO_2$, pupillary findings, secretions
- Treatment: Oxygen, suction, atropine (dose, route, time), decontamination
- Allergies: Medication or chemical allergies
- Medications: Any prehospital or known prescribed meds
- Background: Medical/psychiatric history, similar past events
- Other: ALS or TeleEMS activation, police involvement, environmental risk, crew PPE

## Key Performance Indicators (KPIs)

1. Request ALS back-up within 5 minutes for symptomatic patients.
2. Administer atropine to patients with bronchial secretions or bradycardia.

## References

1. DCAS. (2024). *Clinical Practice Guidelines – Version 02*. Dubai Corporation for Ambulance Services.
2. DCAS. (2024). *Request for Back-Up Policy (DCAS-AOS-QM-PL02-AR)*.
3. DCAS. (2024). *Patient Assessment & Reassessment Policy (DCAS/CLIN/050)*.
4. Greaves, I. and Porter, K. (2007). *Oxford Handbook of Pre-Hospital Care*. Oxford: Oxford University Press.
5. Nelson, L.S., Howland, M.A., Lewin, N.A., Hoffman, R.S., Goldfrank, L.R. (2019). *Goldfrank's Toxicologic Emergencies*. 11th ed. New York: McGraw-Hill Education.
6. Eddleston, M., Buckley, N.A., Eyer, P., and Dawson, A.H. (2008). Management of acute organophosphorus pesticide poisoning. *The Lancet*, 371(9612), pp.597–607. https://doi.org/10.1016/S0140-6736(07)61202-1